

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q13-0206

East Bismarck 230 kV
May 16, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0206. The Interconnection Customer proposes a 96.0 MW solar photovoltaic facility interconnecting at the East Bismarck 230 kV substation in the Illinois area. The request was received on February 20, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 151 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,850,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Mason St 500 kV/230 kV Transformer Bank, the Highgate to Herald-Twin Cities 230kV loads to 133.9% of its 518 MVA rating. Project DFAX: 7.6%.

The Laureles to Big Oaks 100kV is overloaded to 133.8% of its 170 MVA emergency rating for the loss of the Jones Chapel to Dekalb 500kV. The project contributes 5.6% DFAX (5.4 MW) to this overload.

The Dekalb 500/1 kV XFMR is overloaded to 133.6% of its 969 MVA emergency rating for the loss of the Richland to Dekalb Substation 500kV. The project contributes 40.9% DFAX (39.2 MW) to this overload.

The Joso to Onancock Ilesel Generation Facility 230kV is overloaded to 129.8% of its 795 MVA emergency rating for the loss of the Beaver Substation to Eufaula 500kV Line. The project contributes 15.7% DFAX (15.0 MW) to this overload.

Adams to Pickford 230kV: 127.8% loading (531 MVA) following the outage of the Trego to Beaver 500kV L; DFAX = 8.8%.

Russell Fork to Robison Park 230kV: 126.6% loading (1,157 MVA) following the outage of the Dekalb to Jones Chapel 500kV; DFAX = 45.7%.

Newport Tie to Post Rock 230kV: 124.2% loading (864 MVA) following the outage of the Beaver to Trego 500kV; DFAX = 14.4%.

Weedpatch Substation to Canal Road 230kV: 124.2% loading (513 MVA) following the outage of the Trego to Beaver 500kV; DFAX = 8.5%.

For the loss of E Methuen to Lonetree 500kV, the James De Young to St. Lawrence 230kV loads to 124.2% of its 578 MVA rating. Project DFAX: 7.4%.

For the loss of Dekalb to Jones Chapel 500kV, the Cuba City to Farmersville 100kV loads to 122.5% of its 129 MVA rating. Project DFAX: 5.6%.

Bluebonnet to Russell Fork 230kV: 121.4% loading (900 MVA) following the outage of the Dekalb to Jones Chapel 500kV; DFAX = 33.6%.

The E Methuen 500/230kV Autotransformer is overloaded to 120.9% of its 981 MVA emergency rating for the loss of the Swan Vly to Trego 500kV line. The project contributes 13.8% DFAX (13.3 MW) to this overload.

Onancock Diesel Generation Facility to Adams 230kV L: 120.7% loading (1,112 MVA) following the outage of the Beaver to Trego 500kV; DFAX = 17.3%.

Post Rck to Myrtle Pt 230kV: 120.2% loading (931 MVA) following the outage of the Beaver to Trego 500kV; DFAX = 13.9%.

Dekalb Substation 230/1kV Transformer Bank: 119.6% loading (1,082 MVA) following the outage of the Dekalb to Richland 500kV L; DFAX = 40.9%.

Richland to Twin County Ranch 230kV: 116.5% loading (543 MVA) following the outage of the Troy Energy Llc to Richland 500kV; DFAX = 20.4%.

For the loss of Eufaula to Beaver 500kV L, the Beaver to Onancock Diesel Generation Facility 230kV loads to 115.8% of its 911 MVA rating. Project DFAX: 11.9%.

The Muskogee to Darlington Rd 100kV L is overloaded to 115.4% of its 179 MVA emergency rating for the loss of the Richland to Dekalb 500kV. The project contributes 7.0% DFAX (6.7 MW) to this overload.

The Gray'S Flat to E Methuen 230kV is overloaded to 113.0% of its 529 MVA emergency rating for the loss of the Madeline to E Methuen 230kV. The project contributes 9.4% DFAX (9.1 MW) to this overload.

For the loss of Trego to Beaver Substation 500kV Line, the Adams to New Departure 230kV loads to 112.9% of its 613 MVA rating. Project DFAX: 8.5%.

For the loss of Beaver to Trego 500kV, the New Departure to Mason St 230kV loads to 112.8% of its 633 MVA rating. Project DFAX: 8.5%.

Gray'S Flat to Baraga Vlg Substation 230kV line: 112.7% loading (519 MVA) following the outage of the Madeline to E Methuen 230kV; DFAX = 9.4%.

For the loss of Beaver to Eufaula 500kV, the Beaver Substation 500 kV/1 kV XFMR loads to 111.5% of its 1,356 MVA rating. Project DFAX: 15.1%.

Beaver 230/1kV TX: 111.5% loading (1,356 MVA) following the outage of the Bever to Eufaula 500kV L; DFAX = 15.1%.

Minnehaha to Napoleon Peaking Sta 230kV L: 111.3% loading (563 MVA) following the outage of the Troy Energy Llc Substation to Richland 500kV; DFAX = 13.7%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Herald-Twin Cities to Highgate 230kV	Rebuild line with higher-capacity conductor	\$43,305,000	2% (\$852,000)
Laureles to Big Oaks 100kV	Rebuild line with higher-capacity conductor	\$3,305,000	2% (\$64,000)
Dekalb 500/1 kV XFMR	Replace transformer with higher-rated unit	\$24,395,000	2% (\$473,000)
Onancock Diesel Generation Facility to Joso 230kV	Rebuild line with higher-capacity conductor	\$32,559,000	2% (\$643,000)
Adams to Pickford 230kV	Rebuild line with higher-capacity conductor	\$33,895,000	2% (\$667,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Russell Fork to Robison Park 230kV	Rebuild line with higher-capacity conductor	\$35,423,000	2% (\$687,000)
Newport Tie to Post Rock 230kV	Rebuild line with higher-capacity conductor	\$27,212,000	2% (\$536,000)
Weedpatch Substation to Canal Road 230kV	Rebuild line with higher-capacity conductor	\$9,506,000	2% (\$188,000)
James De Young to St. Lawrence 230kV	Rebuild line with higher-capacity conductor	\$80,995,000	2% (\$1,578,000)
Cuba City to Farmersville 100kV	Rebuild line with higher-capacity conductor	\$5,954,000	2% (\$116,000)
Bluebonnet to Russell Fork 230kV	Rebuild line with higher-capacity conductor	\$63,731,000	2% (\$1,236,000)
E Methuen 500/230kV Autotransformer	Replace transformer with higher-rated unit	\$31,314,000	2% (\$609,000)
Adams Substation to Onancock Diesel Generation Facility 230kV L	Rebuild line with higher-capacity conductor	\$63,992,000	2% (\$1,259,000)
Post Rock to Myrtle Pt 230kV	Rebuild line with higher-capacity conductor	\$21,104,000	2% (\$416,000)
Dekalb Substation 230/1kV Transformer Bank	Replace transformer with higher-rated unit	\$10,148,000	2% (\$197,000)
Richland to Twin County Ranch 230kV	Rebuild line with higher-capacity conductor	\$62,082,000	2% (\$1,208,000)
Beaver to Onancock Diesel Generation Facility 230kV	Rebuild line with higher-capacity conductor	\$26,124,000	2% (\$518,000)
Muskogee to Darlington Rd 100kV Line	Rebuild line with higher-capacity conductor	\$21,265,000	2% (\$413,000)
Gray'S Flat to East Methuen 230kV	Rebuild line with higher-capacity conductor	\$47,457,000	2% (\$921,000)
Adams to New Departure 230kV	Rebuild line with higher-capacity conductor	\$43,173,000	2% (\$849,000)
New Departure to Mason St 230kV	Rebuild line with higher-capacity conductor	\$80,375,000	2% (\$1,581,000)
Baraga Vlg to Gray'S Flat Substation 230kV	Rebuild line with higher-capacity conductor	\$17,341,000	2% (\$336,000)
Beaver Substation 500 kV/1 kV XFMR	Replace transformer with higher-rated unit	\$26,661,000	2% (\$528,000)
Beaver 230 kV/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$8,512,000	2% (\$169,000)
Napoleon Peaking Sta to Minnehaha 230kV	Rebuild line with higher-capacity conductor	\$41,462,000	2% (\$806,000)
	Total allocated to project		\$16,850,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.